



Exercise-1

Marked questions are recommended for Revision.

PART - I : SUBJECTIVE QUESTIONS

Section (A): Distance formula, section formula, Area of Triangle & polygon, Collinearity, slope

- A-1.** (i) Prove that the points $(2a, 4a)$, $(2a, 6a)$ and $(2a + \sqrt{3}a, 5a)$ are the vertices of an equilateral triangle whose side is $2a$.
 (ii) Find the points which trisect the line segment joining the points $(0, 0)$ and $(9, 12)$.
- A-2.** (i) In what ratio does the point $\left(\frac{1}{2}, 6\right)$ divide the line segment joining the points $(3, 5)$ and $(-7, 9)$?
 (ii) In which ratio $P(2a - 2, 4a - 6)$ divides $Q(2a - 3, 3a - 7)$ and $R(2a, 6a - 4)$.
- A-3.** (i) Find the value of λ such that points $P(1, 2)$, $Q(-2, 3)$ and $R(\lambda + 1, \lambda)$ are not forming a triangle?
 (ii) Find the ratio in which the line segment joining of the points $(1, 2)$ and $(-2, 3)$ is divided by the line $3x + 4y = 7$
 (iii) Find the harmonic conjugate of the point $R(5, 1)$ with respect to points $P(2, 10)$ and $Q(6, -2)$.
- A-4.** A and B are the points $(3, 4)$ and $(5, -2)$ respectively. Find the co-ordinates of a point P such that $PA = PB$ and the area of the triangle $PAB = 10$.
- A-5.** Find the area of the quadrilateral with vertices as the points given in each of the following :
 (i) $(0, 0)$, $(4, 3)$, $(6, 0)$, $(0, 3)$ (ii) $(0, 0)$, $(a, 0)$, (a, b) , $(0, b)$

Section (B) : Different forms of straight lines and Angle between lines

- B-1.** Reduce $x + \sqrt{3}y + 4 = 0$ to the :
 (i) Slope intercepts form and find its slope and y-intercept.
 (ii) Intercepts form and find its intercepts on the axes.
 (iii) Normal form and find values of P and α .
- B-2.** Find number of straight line passing through $(2, 4)$ and forming a triangle of 16 sq. cm with the coordinate axis.
- B-3.** Find the equation of the straight line that passes through the point $A(-5, -4)$ and is such that the portion intercepted between the axes is divided by the point A in the ratio 1 : 2 (internally).
- B-4.** The co-ordinates of the mid-points of the sides of a triangle are $(2, 1)$, $(5, 3)$ and $(3, 7)$. Find the length and equation of its sides.
- B-5.** Find the straight line cutting an intercept of one unit on negative x-axis and inclined at 45° (in anticlockwise direction) with positive direction of x-axis
- B-6.** Through the point $P(4, 1)$ a line is drawn to meet the line $3x - y = 0$ at Q where $PQ = \frac{11}{2\sqrt{2}}$. Determine the equation of line.
- B-7.** A line having slope '1' is drawn from a point $A(-3, 0)$ cuts a curve $y = x^2 + x + 1$ at P & Q. Find $\ell(AP)$ $\ell(AQ)$.





- B-8.** Find the direction in which a straight line must be drawn through the point (1, 2), so that its point of intersection with the line $x + y = 4$ may be at a distance $\frac{\sqrt{6}}{3}$ from this point.
- B-9.** Through the point (3, 4) are drawn two straight lines each inclined at 45° to the straight line $x - y = 2$. Find their equations and also find the area of triangle bounded by the three lines.
- B-10.** Find the equation of a straight line which passes through the point (2, 1) and makes an angle of $\pi/4$ with the straight line $2x + 3y + 4 = 0$
- B-11.** From (1, 4) you travel $5\sqrt{2}$ units by making 135° angle with positive x-axis (anticlockwise) and then 4 units by making 120° angle with positive x-axis (clockwise) to reach Q. Find co-ordinates of point Q.
- B-12.** The ends of the hypotenuse of a right angled triangle are (6, 0) and (0, 6), then find the locus of third vertex of triangle.
- B-13.** A point moves in the x-y plane such that the sum of its distances from two mutually perpendicular lines is always equal to 3, then find the area enclosed by the locus of the point.
- B-14.** One side of a rectangle lies along the line $4x + 7y + 5 = 0$. Two of its vertices are (-3, 1) and (1, 1). Then find the equations of other sides.

Section (C) : Position of point, linear inequation, perpendicular distance, image & foot, Area of Parallelogram

- C-1.** Plot the region
(i) $6x + 2y \geq 31$ (ii) $2x + 5y \leq 10$ (iii) $8x + 3y + 6 > 0$ (iv) $x > 2$
- C-2.** Find coordinates of the foot of perpendicular, image and equation of perpendicular drawn from the point (2, 3) to the line $y = 3x - 4$.
- C-3.** Starting at the origin, a beam of light hits a mirror (in the form of a line) at the point A(4, 8) and reflected line passes through the point B (8, 12). Compute the slope of the mirror.
- C-4.** Find the nearest point on the line $3x + 4y - 1 = 0$ from the origin.
- C-5.** Find the position of the origin with respect to the triangle whose sides are $x + 1 = 0$, $3x - 4y - 5 = 0$, $5x + 12y - 27 = 0$.
- C-6.** Find the area of parallelogram whose two sides are $y = x + 3$, $2x - y + 1 = 0$ and remaining two sides are passing through (0, 0).
- C-7.** Is there a real value of λ for which the image of the point $(\lambda, \lambda - 1)$ by the line mirror $3x + y = 6\lambda$ is the point $(\lambda^2 + 1, \lambda)$? If so, find λ .
- C-8.** Find the equations of two straight lines which are parallel to $x + 7y + 2 = 0$ and at $\sqrt{2}$ distance away from it.
- C-9.** Prove that the area of the parallelogram contained by the lines $4y - 3x - a = 0$, $3y - 4x + a = 0$, $4y - 3x - 3a = 0$ and $3y - 4x + 2a = 0$ is $\frac{2}{7} a^2$.

Section (D) : Centroid, orthocentre, circumcentre, incentre, excentre, Locus

- D-1.** For triangle whose vertices are (0, 0), (5, 12) and (16, 12). Find coordinates of
(i) Centroid (ii) Circumcentre



- (iii) Incentre (iv) Excentre opposite to vertex (5, 12)
- D-2.** Find the sum of coordinates of the orthocentre of the triangle whose sides are $x = 3$, $y = 4$ and $3x + 4y = 6$.
- D-3.** Find equations of altitudes and the co-ordinates of the orthocentre of the triangle whose sides are $3x - 2y = 6$, $3x + 4y + 12 = 0$ and $3x - 8y + 12 = 0$.
- D-4.** A triangle has the lines $y = m_1x$ and $y = m_2x$ for two of its sides, where m_1, m_2 are the roots of the equation $x^2 + ax - 1 = 0$, then find the orthocentre of triangle.
- D-5.** Prove that the circumcentre, orthocentre, incentre & centroid of the triangle formed by the points $A(-1, 11)$; $B(-9, -8)$; $C(15, -2)$ are collinear, without actually finding any of them.
- D-6.** Find locus of centroid of $\triangle AOB$ if line AB passes through $(3, 2)$, A and B are on coordinate axes.
- D-7.** Find the locus of the centroid of a triangle whose vertices are $(a \cos t, a \sin t)$, $(b \sin t, -b \cos t)$ and $(1, 0)$, where 't' is the parameter.
- D-8.** Show that equation of the locus of a point which moves so that difference of its distance from two given points $(ae, 0)$ and $(-ae, 0)$ is equal to $2a$ is $\frac{x^2}{a^2} - \frac{y^2}{a^2(e^2 - 1)} = 1$.
- D-9.** Find the locus of point of intersection of the lines $x \cos \alpha + y \sin \alpha = a$ and $x \sin \alpha - y \cos \alpha = b$, where α is a parameter.
- D-10.** The area of the triangle formed by the intersection of a line parallel to x-axis and passing through $P(h, k)$ with the lines $y = x$ and $x + y = 2$ is $4h^2$. Find the locus of the point P .

Section (E) : Angle Bisector, condition of concurrency, family of straight lines

- E-1.** Find equations of acute and obtuse angle bisectors of the angle between the lines $4x + 3y - 7 = 0$ and $24x + 7y - 31 = 0$.
- E-2.** Find the equation of a straight line passing through the point $(4, 5)$ and equally inclined to the lines $3x = 4y + 7$ and $5y = 12x + 6$.
- E-3.** The line $x + 3y - 2 = 0$ bisects the angle between a pair of straight lines of which one has equation $x - 7y + 5 = 0$, then find equation of other line
- E-4.** Find the value of λ such that lines $x + 2y = 3$, $3x - y = 1$ and $\lambda x + y = 2$ can not form a triangle.
- E-5.** Find values of λ for which line $y = x + 1$, $y = \lambda x + 2$ and $y = (\lambda^2 + \lambda - 1)x + 3$ are con-current.
- E-6.** Find the equation to the straight line passing through
 (i) The point $(3, 2)$ and the point of intersection of the lines $2x + 3y = 1$ and $3x - 4y = 6$.
 (ii) The intersection of the lines $x + 2y + 3 = 0$ and $3x + 4y + 7 = 0$ and perpendicular to the straight line $y - x = 8$.
- E-7.** Find the locus of the circumcentre of a triangle whose two sides are along the co-ordinate axes and third side passes through the point of intersection of the lines $ax + by + c = 0$ and $\ell x + my + n = 0$.

Section (F) : Pair of straight lines, Homogenization

- F-1.** If the slope of one of the lines represented by $ax^2 + 2hxy + by^2 = 0$ be the n^{th} power of the other, then prove that $(ab^n)^{\frac{1}{n+1}} + (a^n b)^{\frac{1}{n+1}} + 2h = 0$



- F-2.** For what value of λ does the equation $12x^2 - 10xy + 2y^2 + 11x - 5y + \lambda = 0$ represent a pair of straight lines? Find their
 (i) equations, (ii) point of intersection, (iii) acute angle between them.
- F-3.** (i) Find the integral values of 'h' for which $hx^2 - 5xy + 4hy^2 + x + 2y - 2 = 0$ represents two real straight lines.
 (ii) If the pair of lines represented by equation $k(k - 3)x^2 + 16xy + (k + 1)y^2 = 0$ are perpendicular to each other, then find k.
- F-4.** Find the equation of the straight lines joining the origin to the points of intersection of the line $lx + my + n = 0$ and the curve $y^2 = 4ax$. Also, find the condition of their perpendicularity.
- F-5.** Find the condition that the diagonals of the parallelogram formed by the lines $ax + by + c = 0$; $ax + by + c' = 0$; $a'x + b'y + c = 0$, $a'x + b'y + c' = 0$ are at right angles. Also find the equation to the diagonals of the parallelogram.

PART - II : ONLY ONE OPTION CORRECT TYPE

Section (A): Distance formula, section formula, Area of Triangle & polygon, Collinearity, slope

- A-1.** Mid point of A(0, 0) and B(1024, 2048) is A_1 . mid point of A_1 and B is A_2 and so on. Coordinates of A_{10} are.
 (A) (1022, 2044) (B) (1025, 2050) (C) (1023, 2046) (D) (1, 2)
- A-2.** If the points $(k, 2 - 2k)$, $(1 - k, 2k)$ and $(-k - 4, 6 - 2k)$ be collinear, the number of possible values of k are
 (A) 4 (B) 2 (C) 1 (D) 3
- A-3.** Given a $\triangle ABC$ with unequal sides. P is the set of all points which is equidistant from B & C and Q is the set of all point which is equidistant from sides AB and AC. Then $n(P \cap Q)$ equals :
 (A) 1 (B) 2 (C) 3 (D) Infinite
- A-4.** A line segment AB is divided internally and externally in the same ratio (> 1) at P and Q respectively and M is mid point of AB.
Statement-1: MP, MB, MQ are in G.P.
Statement-2: AP, AB and AQ are in H.P.
 (A) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is correct explanation for STATEMENT-1
 (B) STATEMENT-1 is true, STATEMENT-2 is true and STATEMENT-2 is not correct explanation for STATEMENT-1
 (C) STATEMENT-1 is true, STATEMENT-2 is false
 (D) STATEMENT-1 is false, STATEMENT-2 is true
 (E) Both STATEMENTS are false
- A-5.** Find the area of the triangle formed by the mid points of sides of the triangle whose vertices are (2, 1), (-2, 3), (4, -3)
 (A) 1.5 sq. units (B) 3 sq. units (C) 6 sq. units (D) 12 sq. units

Section (B) : Different forms of straight lines and Angle between lines

- B-1.** A straight line through P (1, 2) is such that its intercept between the axes is bisected at P. Its equation is :
 (A) $x + 2y = 5$ (B) $x - y + 1 = 0$ (C) $x + y - 3 = 0$ (D) $2x + y - 4 = 0$
- B-2.** The number of integral points (integral point means both the coordinates should be integer) exactly in the interior of the triangle with vertices (0, 0), (0, 21) and (21, 0), is
 (A) 133 (B) 190 (C) 233 (D) 105





- B-3.** The line joining two points A (2, 0) and B (3, 1) is rotated about A in the anticlockwise direction through an angle of 15° . The equation of the line in the new position is :
 (A) $x - \sqrt{3}y - 2 = 0$ (B) $x - 2y - 2 = 0$ (C) $\sqrt{3}x - y - 2\sqrt{3} = 0$ (D) $\sqrt{2}x - y - 2\sqrt{2} = 0$
- B-4.** In a $\triangle ABC$, side AB has the equation $2x + 3y = 29$ and the side AC has the equation $x + 2y = 16$. If the mid point of BC is (5, 6), then the equation of BC is
 (A) $2x + y = 16$ (B) $x + y = 11$ (C) $2x - y = 4$ (D) $x + y = 10$
- B-5.** A square of side 'a' lies above the x-axis and has one vertex at the origin. The side passing through the origin makes an angle α ($0 < \alpha < \frac{\pi}{4}$) with the positive direction of x-axis. The equation of its diagonal not passing through the origin is :
 (A) $y(\cos \alpha - \sin \alpha) - x(\sin \alpha - \cos \alpha) = a$ (B) $y(\cos \alpha + \sin \alpha) + x(\sin \alpha - \cos \alpha) = a$
 (C) $y(\cos \alpha + \sin \alpha) + x(\sin \alpha + \cos \alpha) = a$ (D) $y(\cos \alpha + \sin \alpha) + x(\cos \alpha - \sin \alpha) = a$
- B-6.** Find equation of a straight line on which length of perpendicular from the origin is four units and the line makes an angle of 120° with the positive direction of x-axis.
 (A) $\sqrt{3}x - y = 0$ (B) $\sqrt{3}x + y = 8$ (C) $x + \sqrt{3}y = 8$ (D) $x - \sqrt{3}y = 8$
- B-7.** The distance of the point (2, 3) from the line $2x - 3y + 9 = 0$ measured along a line $x - y + 1 = 0$ is :
 (A) $5\sqrt{3}$ (B) $4\sqrt{2}$ (C) $3\sqrt{2}$ (D) $2\sqrt{2}$
- B-8.** If a point P(x, y) from where line drawn cuts coordinate axes at A and B (with A on x-axis and B on y-axis) satisfies $\alpha \cdot \frac{x^2}{PB^2} + \beta \cdot \frac{y^2}{PA^2} = 1$, then $\alpha + \beta$ is
 (A) 1 (B) 2 (C) 3 (D) 4
- B-9.** Two particles start from the point (2, -1), one moving 2 units along the line $x + y = 1$ and the other 5 units along the line $x - 2y = 4$. If the particles move towards increasing y, then their new positions are
 (A) $(2 - \sqrt{2}, \sqrt{2} - 1)$, $(2\sqrt{5} + 2, \sqrt{5} - 1)$ (B) $(2\sqrt{5} + 2, \sqrt{5} - 1)$, $(2\sqrt{2}, \sqrt{2} + 1)$
 (C) $(2 + \sqrt{2}, \sqrt{2} + 1)$, $(2\sqrt{5} + 2, \sqrt{5} + 1)$ (D) none of these
- B-10.** Equation of a straight line passing through the origin and making with x - axis an acute angle twice the size of the angle made by the line $y = (0.2)x$ with the x - axis, is :
 (A) $y = (0.4)x$ (B) $y = (5/12)x$ (C) $6y - 5x = 0$ (D) $6y + 5x = 0$
- B-11.** The points A (1, 3) and C (5, 1) are the opposite vertices of rectangle. The equation of line passing through other two vertices and of gradient 2, is
 (A) $2x + y - 8 = 0$ (B) $2x - y - 4 = 0$ (C) $2x - y + 4 = 0$ (D) $2x + y = 0$
- B-12.** The point (-4, 5) is the vertex of a square and one of its diagonals is $7x - y + 8 = 0$. The equation of the other diagonal is
 (A) $7x - y + 23 = 0$ (B) $7y + x = 30$ (C) $7y + x = 31$ (D) $x - 7y = 30$
- B-13.** Two straight lines $x + 2y = 2$ and $x + 2y = 6$ are given, then find the equation of the line parallel to given lines and divided distance between lines in the ratio 2 : 1 internally
 (A) $3x + 6y + 8 = 0$ (B) $3x + 6y = 14$ (C) $3x + 6y + 14 = 0$ (D) $3x + 2y = 10$

Section (C) : Position of point, linear inequation, perpendicular distance, image & foot

- C-1.** The set of values of 'b' for which the origin and the point (1, 1) lie on the same side of the straight line, $a^2x + aby + 1 = 0 \quad \forall \quad a \in \mathbb{R}, b > 0$ are :
 (A) $b \in (2, 4)$ (B) $b \in (0, 2)$ (C) $b \in [0, 2]$ (D) $(2, \infty)$



- C-2.** The point $(a^2, a + 1)$ is a point in the angle between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin, then
 (A) $a \geq 1$ or $a \leq -3$ (B) $a \in (-3, 0) \cup (1/3, 1)$
 (C) $a \in (0, 1)$ (D) $a \in (-\infty, 0)$
- C-3.** Find area of region represented by $3x + 4y > 12$, $4x + 3y > 12$ and $x + y < 4$
 (A) $\frac{8}{7}$ (B) $\frac{4}{7}$ (C) $\frac{7}{8}$ (D) $\frac{8}{7}$
- C-4.** The image of the point A $(1, 2)$ by the line mirror $y = x$ is the point B and the image of B by the line mirror $y = 0$ is the point (α, β) , then :
 (A) $\alpha = 1, \beta = -2$ (B) $\alpha = 0, \beta = 0$ (C) $\alpha = 2, \beta = -1$ (D) $\alpha = 1, \beta = -1$
- C-5.** The equations of the perpendicular bisector of the sides AB and AC of a $\triangle ABC$ are $x - y + 5 = 0$ and $x + 2y = 0$ respectively. If the point A is $(1, -2)$, then the equation of the line BC is :
 (A) $14x + 23y = 40$ (B) $14x - 23y = 40$ (C) $23x + 14y = 40$ (D) $23x - 14y = 40$
- C-6.** A light beam emanating from the point A $(3, 10)$ reflects from the straight line $2x + y - 6 = 0$ and then passes through the point B $(4, 3)$. The equation of the reflected beam is $x + 3y - \lambda = 0$, then the value of λ is
 (A) 11 (B) 12 (C) 13 (D) 14

Section (D) : Centroid, orthocentre, circumcentre, incentre, excentre, Locus

- D-1.** The orthocentre of the triangle ABC is 'B' and the circumcentre is 'S' (a, b) . If A is the origin, then the co-ordinates of C are :
 (A) $(2a, 2b)$ (B) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (C) $\left(\sqrt{a^2 + b^2}, 0\right)$ (D) none
- D-2.** A triangle ABC with vertices A $(-1, 0)$, B $(-2, 3/4)$ & C $(-3, -7/6)$ has its orthocentre H. Then the orthocentre of triangle BCH will be :
 (A) $(-3, -2)$ (B) $(1, 3)$ (C) $(-1, 2)$ (D) none of these
- D-3.** Find locus of centroid of $\triangle ABC$, if B $(1, 1)$, C $(4, 2)$ and A lies on the line $y = x + 3$.
 (A) $3x + 3y + 1 = 0$ (B) $x + y = 3$ (C) $3x - 3y + 1 = 0$ (D) $x - y = 3$
- D-4.** The locus of the mid-point of the distance between the axes of the variable line $x \cos \alpha + y \sin \alpha = p$, where p is constant, is
 (A) $x^2 + y^2 = 4p^2$ (B) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{p^2}$ (C) $x^2 + y^2 = \frac{4}{p^2}$ (D) $\frac{1}{x^2} - \frac{1}{y^2} = \frac{2}{p^2}$
- D-5.** Find the locus of a point which moves so that sum of the squares of its distance from the axes is equal to 3.
 (A) $x^2 + y^2 = 9$ (B) $x^2 + y^2 = 3$ (C) $|x| + |y| = 3$ (D) $x^2 - y^2 = 3$
- D-6.** A variable straight line passes through a fixed point (a, b) intersecting the co-ordinates axes at A & B. If 'O' is the origin, then the locus of the centroid of the triangle OAB is :
 (A) $bx + ay - 3xy = 0$ (B) $bx + ay - 2xy = 0$ (C) $ax + by - 3xy = 0$ (D) $ax + by - 2xy = 0$
- D-7.** Consider a triangle ABC, whose vertices are A $(-2, 1)$, B $(1, 3)$ and C (x, y) . If C is a moving point such that area of $\triangle ABC$ is constant, then locus of C is :
 (A) Straight line (B) Circle (C) Ray (D) Parabola



- D-8.** If the equation of the locus of a point equidistant from the points (a_1, b_1) and (a_2, b_2) is $(a_1 - a_2)x + (b_1 - b_2)y + c = 0$, then the value of 'c' is :
- (A) $\frac{1}{2} (a_2^2 + b_2^2 - a_1^2 - b_1^2)$ (B) $a_1^2 - a_2^2 + b_1^2 - b_2^2$
 (C) $\frac{1}{2} (a_1^2 + a_2^2 + b_1^2 + b_2^2)$ (D) $\sqrt{a_1^2 + b_1^2 - a_2^2 - b_2^2}$

Section (E) : Angle Bisector, condition of concurrency, family of straight lines

- E-1.** The equation of bisectors of two lines L_1 & L_2 are $2x - 16y - 5 = 0$ and $64x + 8y + 35 = 0$. If the line L_1 passes through $(-11, 4)$, the equation of acute angle bisector of L_1 & L_2 is :
 (A) $2x - 16y - 5 = 0$ (B) $64x + 8y + 35 = 0$ (C) $2x + 16y + 5 = 0$ (D) $2x + 16y - 5 = 0$
- E-2.** The equation of the internal bisector of $\angle BAC$ of $\triangle ABC$ with vertices $A(5, 2)$, $B(2, 3)$ and $C(6, 5)$ is
 (A) $2x + y + 12 = 0$ (B) $x + 2y - 12 = 0$ (C) $2x + y - 12 = 0$ (D) $2x - y - 12 = 0$
- E-3.** The equation of the bisector of the angle between two lines $3x - 4y + 12 = 0$ and $12x - 5y + 7 = 0$ which contains the point $(-1, 4)$ is :
 (A) $21x + 27y - 121 = 0$ (B) $21x - 27y + 121 = 0$
 (C) $21x + 27y + 191 = 0$ (D) $\frac{-3x + 4y - 12}{5} = \frac{12x - 5y + 7}{13}$
- E-4.** The least positive value of t so that the lines $x = t + a$, $y + 16 = 0$ and $y = ax$ (where a is real variable) are concurrent is
 (A) 2 (B) 4 (C) 16 (D) 8
- E-5.** Consider the family of lines $5x + 3y - 2 + \lambda_1(3x - y - 4) = 0$ and $x - y + 1 + \lambda_2(2x - y - 2) = 0$. Equation of a straight line that belong to both families is -
 (A) $25x - 62y + 86 = 0$ (B) $62x - 25y + 86 = 0$ (C) $25x - 62y = 86$ (D) $5x - 2y - 7 = 0$
- E-6.** The equation of a line of the system $2x + y + 4 + \lambda(x - 2y - 3) = 0$ which is at a distance $\sqrt{10}$ units from point $A(2, -3)$ is
 (A) $3x + y + 1 = 0$ (B) $3x - y + 1 = 0$ (C) $y - 3x + 1 = 0$ (D) $y + 3x - 2 = 0$
- E-7.** The lines $ax + by + c = 0$, where $3a + 2b + 4c = 0$, are concurrent at the point
 (A) $\left(\frac{1}{2}, \frac{3}{4}\right)$ (B) $(1, 3)$ (C) $(3, 1)$ (D) $\left(\frac{3}{4}, \frac{1}{2}\right)$
- E-8.** Find the equation of a straight line which passes through the point of intersection of the straight lines $x + y - 5 = 0$ and $x - y + 3 = 0$ and perpendicular to the straight line intersecting x -axis at the point $(-2, 0)$ and the y -axis at the point $(0, -3)$,
 (A) $2x + 3y + 10 = 0$ (B) $2x - 3y + 10 = 0$ (C) $2x - 5y + 10 = 0$ (D) $2x + 5y + 10 = 0$
- E-9.** The line parallel to the x -axis and passing through the intersection of the lines $ax + 2by + 3b = 0$ and $bx - 2ay - 3a = 0$, where $(a, b) \neq (0, 0)$ is
 (A) Above the x -axis at a distance of $3/2$ from it (B) Above the x -axis at a distance of $2/3$ from it
 (C) Below the x -axis at a distance of $3/2$ from it (D) Below the x -axis at a distance of $2/3$ from it

Section (F) : Pair of straight lines, Homogenization

- F-1.** If the slope of one line of the pair of lines represented by $ax^2 + 10xy + y^2 = 0$ is four times the slope of the other line, then $a =$
 (A) 1 (B) 2 (C) 4 (D) 16



- F-2.** The equation of second degree $x^2 + 2\sqrt{2}xy + 2y^2 + 4x + 4\sqrt{2}y + 1 = 0$ represents a pair of straight lines. The distance between them is
 (A) 4 (B) $\frac{4}{\sqrt{3}}$ (C) 2 (D) $2\sqrt{3}$
- F-3.** The straight lines joining the origin to the points of intersection of the line $2x + y = 1$ and curve $3x^2 + 4xy - 4x + 1 = 0$ include an angle :
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

PART - III : MATCH THE COLUMN

- 1. Column - I**
- (A) The points A(-4, -1), B (-2, -4), C (4, 0) and D(2, 3) are the vertices of
- (B) The figure formed by the lines $2x + 5y + 4 = 0$, $5x + 2y + 7 = 0$, $2x + 5y + 3 = 0$ and $5x + 2y + 6 = 0$ is
- (C) A quadrilateral whose diagonal are perpendicular bisectors of each other must be
- (D) A quadrilateral whose diagonals are angle bisector of sides and bisect each other must be
- Column - II**
- (p) Square
- (q) Rectangle
- (r) Rhombus
- (s) Parallelogram
- 2. Match The Followings :**
- Column - I**
- (A) P lies on the line $y = x$ and Q lies on $y = 2x$. The equation for the locus of the mid point of PQ, if $|PQ| = 4$, is $25x^2 - \lambda xy + 13y^2 = 4$, then λ equals
- (B) The line $(K + 1)^2 x + ky - 2K^2 - 2 = 0$ passes through a point (α, β) regardless of value K. Then $(\alpha - \beta)$ equals :
- (C) Let two lines be $C_1 : 3x - 4y + 1 = 0$ and $C_2 : 8x + 6y + 1 = 0$ suppose (α, β) is a point which is at unit distance from each of the given lines. then sum of all possible value of α is
- (D) If distance between the pair of parallel lines $x^2 + 2xy + y^2 - 8ax - 8ay - 9a^2 = 0$ is $25\sqrt{2}$, then 'a/5' is equal to
- Column - II**
- (p) 36
- (q) 6
- (r) -4/5
- (s) -1

Exercise-2

Marked questions are recommended for Revision.

PART - I : ONLY ONE OPTION CORRECT TYPE

1. On the portion of the straight line $x + 2y = 4$ intercepted between the axes, a square is constructed on the side of the line away from the origin. Then the point of intersection of its diagonals has co-ordinates
 (A) (2, 3) (B) (3, 2) (C) (3, 3) (D) (2, 2)
2. If the straight line $ax + by + p = 0$ and $x\cos\alpha + y\sin\alpha = p$ enclosed an angle of $\frac{\pi}{4}$ and the line $x\sin\alpha - y\cos\alpha = 0$ meets them at the same point, then $a^2 + b^2$ is
 (A) 4 (B) 3 (C) 2 (D) 1



3. A ΔABC is formed by the lines $2x - 3y - 6 = 0$, $3x - y + 3 = 0$ and $3x + 4y - 12 = 0$. If the points $P(\alpha, 0)$ and $Q(0, \beta)$ always lie on or inside the ΔABC , then ;
 (A) $\alpha \in [-1, 2]$ & $\beta \in [-2, 3]$ (B) $\alpha \in [-1, 3]$ & $\beta \in [-2, 4]$
 (C) $\alpha \in [-2, 4]$ & $\beta \in [-3, 4]$ (D) $\alpha \in [-1, 3]$ & $\beta \in [-2, 3]$
4. If $P\left(1 + \frac{t}{\sqrt{2}}, 2 + \frac{t}{\sqrt{2}}\right)$ be any point on a line, then the range of values of t for which the point P lies between the parallel lines $x + 2y = 1$ and $2x + 4y = 15$ is
 (A) $-\frac{4\sqrt{2}}{3} < t < \frac{5\sqrt{2}}{6}$ (B) $0 < t < \frac{5\sqrt{2}}{6}$ (C) $-\frac{4\sqrt{2}}{5} < t < 0$ (D) $-\frac{4\sqrt{2}}{3} < t < \frac{\sqrt{2}}{6}$
5. The point $A(4, 1)$ undergoes following transformations successively :
 (i) reflection about line $y = x$
 (ii) translation through a distance of 3 units in the positive direction of x -axis
 (iii) rotation through an angle 105° in anti-clockwise direction about origin O .
 Then the final position of point A is
 (A) $\left(\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$ (B) $(-2, 7\sqrt{2})$ (C) $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$ (D) $(-2\sqrt{6}, 2\sqrt{2})$
6. Given two points $A \equiv (-2, 0)$ and $B \equiv (0, 4)$, then find coordinate of a point P lying on the line $2x - 3y = 9$ so that perimeter of ΔAPB is least.
 (A) $\left(\frac{42}{13}, -\frac{11}{3}\right)$ (B) $\left(\frac{84}{13}, -\frac{74}{13}\right)$ (C) $\left(\frac{21}{17}, -\frac{37}{17}\right)$ (D) $(0, -3)$
7. A ray of light is sent from the point $(1, 4)$. Upon reaching the x -axis, the ray is reflected from the point $(3, 0)$. This reflected ray is again reflected by the line $x + y = 5$ and intersect y -axis at P . Find the co-ordinate of P .
 (A) $\left(\frac{1}{2}, 0\right)$ (B) $\left(0, -\frac{1}{2}\right)$ (C) $\left(0, \frac{1}{3}\right)$ (D) $\left(2, \frac{1}{-2}\right)$
8. AB is a variable line sliding between the co-ordinate axes in such a way that A lies on X -axis and B lies on Y -axis. If P is a variable point on AB such that $PA = b$, $PB = a$ and $AB = a + b$, then equation of locus of P is
 (A) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (B) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (C) $x^2 + y^2 = a^2 + b^2$ (D) $x^2 - y^2 = a^2 + b^2$
9. If the distance of any point (x, y) from the origin is defined as $d(x, y) = \max \{ |x|, |y| \}$, $d(x, y) = a$ (where 'a' is non-zero constant), then the locus is
 (A) A circle (B) Straight line (C) A square (D) A triangle
10. Two ends A & B of a straight line segment of constant length 'c' slide upon the fixed rectangular axes OX & OY respectively. If the rectangle $OAPB$ is completed. Then find locus of the foot of the perpendicular drawn from P to AB .
 (A) $x^{2/3} + y^{2/3} = c^{2/3}$ (B) $x^{2/3} + y^{2/3} = c^{1/3}$ (C) $x^{1/3} + y^{1/3} = c^{2/3}$ (D) $x^{1/3} + y^{1/3} = c^{1/3}$
11. Let the line $\frac{x}{a} + \frac{y}{b} = 1$ cuts the x and y axes at A and B respectively. Now a line parallel to the given line cuts the coordinate axis at P and Q and points P and Q are joined to B and A respectively. The locus of intersection of the joining lines is
 (A) $\frac{x}{a} - \frac{y}{b} = 0$ (B) $\frac{x}{a} + \frac{y}{b} = 0$ (C) $\frac{x}{b} - \frac{y}{a} = 0$ (D) $\frac{x}{b} + \frac{y}{a} = 0$





12. A variable line whose slope is -2 cuts the x and y axes respectively at points A and C . A rhombus $ABCD$ is completed such that vertex B lies on the line $y = x$. Then the locus of vertex D is
 (A) $2x + y = 1$ (B) $x - y = 0$ (C) $x + y = 0$ (D) $x + 2y = 0$
13. $ABCD$ is a square away from origin of side length ' a '. Its side AB slides between x and y -axes in first quadrant with A on x -axis and B on y -axis. The locus of the foot of perpendicular dropped from the point E on the diagonal AC (where E is the midpoint of the side AD), is
 (A) $(y - x)^2 + (x - 3y)^2 = a^2$ (B) $(y - x)^2 + (x - 3y)^2 = \frac{a^2}{2}$
 (C) $(y - x)^2 + (x - 3y)^2 = \frac{a^2}{4}$ (D) None of these
14. Let orthocentre of $\triangle ABC$ is $(4, 6)$. If $A = (4, 7)$ and $B = (-2, 4)$, then find coordinates of vertex C
 (A) $(5, 4)$ (B) $(4, 5)$ (C) $(-5, -4)$ (D) $(-4, -5)$
15. Let Incentre of $\triangle ABC$ is $(2, 5)$. If $A = (1, 13)$ and $B = (-4, 1)$, then find coordinates of C
 (A) $(1, 10)$ (B) $(10, 1)$ (C) $(8, 2)$ (D) $(9, 3)$
16. The locus of circumcentre of the triangle formed by vertices $A((-pq - p - q), -(1 + p)(1 + q))$, $B(pq + p - q, (1 + p)(1 + q))$, $C(pq + q - p, (1 + p)(1 + q))$ is
 (A) $y + x = 0$ (B) $y - x = 0$ (C) $x^2 + y^2 = 1$ (D) $xy = 1$
17. Let two sides of rectangle of area 20 units are along lines $x - y = 0$ and $x + y = 2$, then the locus of point of intersection of diagonals is
 (A) $(x - 1)^2 + (y - 1)^2 = 10$ or $(y - 1)^2 + (x - 1)^2 = 10$
 (B) $(x - 1)^2 - (y - 1)^2 = 10$ or $(y - 1)^2 - (x - 1)^2 = 10$
 (C) $(x + 1)^2 - (y + 1)^2 = 10$ or $(y + 1)^2 - (x + 1)^2 = 10$
 (D) $(x - 1)^2 + (y - 1)^2 = 10$ or $(y + 1)^2 - (x + 1)^2 = 10$
18. Area of the triangle formed by the line $x + y = 3$ and angle bisectors of the pair of straight lines $x^2 - y^2 + 2y = 1$ is
 (A) 2 sq units (B) 4 sq. units (C) 6 sq. units (D) 8 sq. units
19. Equation of the line pair through the origin and perpendicular to the line pair $xy - 3y^2 + y - 2x + 10 = 0$ is
 (A) $xy - 3y^2 = 0$ (B) $xy + 3x^2 = 0$ (C) $xy + 3y^2 = 0$ (D) $x^2 - y^2 = 0$
20. Find the equation of the two straight lines which together with those given by the equation $6x^2 - xy - y^2 + x + 12y - 35 = 0$ will make a parallelogram whose diagonals intersect in the origin.
 (A) $6x^2 - xy - y^2 - x - 12y - 35 = 0$ (B) $6x^2 - xy - y^2 - x - 12y + 35 = 0$
 (C) $6x^2 - xy - y^2 - x + 12y - 35 = 0$ (D) $6x^2 - xy - y^2 + x - 12y - 35 = 0$
21. The curve passing through the points of intersection of $S_1 \equiv \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0$ and $S_2 \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines which are
 (A) equally inclined to the x - axis (B) perpendicular to each other
 (C) parallel to each other (D) Not equally inclined to y -axis



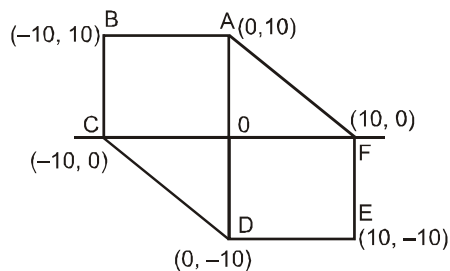
PART-II: NUMERICAL VALUE QUESTIONS

INSTRUCTION :

- ❖ The answer to each question is **NUMERICAL VALUE** with two digit integer and decimal upto two digit.
- ❖ If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal placed.

1. If the points $\left(\frac{a^3}{a-1}, \frac{a^2-3}{a-1}\right)$, $\left(\frac{b^3}{b-1}, \frac{b^2-3}{b-1}\right)$ and $\left(\frac{c^3}{c-1}, \frac{c^2-3}{c-1}\right)$ are collinear for three distinct values a, b, c and $a \neq 1$, $b \neq 1$ and $c \neq 1$, then find the value of $abc - (ab + bc + ac) + 3(a + b + c)$.
2. The complete set of values of λ for which point $(\lambda, \lambda + 1)$ is an interior points of $\triangle ABC$, where $A \equiv (0, 3)$, $B \equiv (-2, 0)$ and $C \equiv (6, 1)$ is (p, q) then value of $p^2 + q^2$ is
3. Let ABC be a triangle such that the coordinates of the vertex A are $(-3, 1)$. Equation of the median through B is $2x + y - 3 = 0$ and equation of the angular bisector of C is $7x - 4y - 1 = 0$. Find the slope of line BC.
4. $A(3, 4)$, $B(0, 0)$ and $C(3, 0)$ are vertices of $\triangle ABC$. If 'P' is the point inside the $\triangle ABC$, such that $d(P, BC) \leq \min. \{d(P, AB), d(P, AC)\}$. Then the maximum of $d(P, BC)$ is. (where $d(P, BC)$ represent distance between P and BC).
5. Drawn from the origin are two mutually perpendicular straight lines forming an isosceles triangle together with the straight line $2x + y = 13$. Then find the area of the triangle.
6. On the straight line $y = x + 2$, a point (a, b) is such that the sum of the square of distances from the straight lines $3x - 4y + 8 = 0$ and $3x - y - 1 = 0$ is least, then find value of $(a + b)$.
7. Parallelogram ABCD is cut by $(2n - 1)$ number of parallel lines in which one is diagonal AC. Distance between any two nearest lines is same which is also equal to distance of B, D from respective nearest line among these. Ratio of area of smallest triangle so formed to area of parallelogram is $\frac{1}{32}$. Find n.
8. A is a variable point on x-axis and $B(0, b)$ is a fixed point. A equilateral triangle ABC is completed with vertex C away from origin. If the locus of the point C is $\alpha x + \beta y = b$, then $\alpha^3 + \beta^3$ is
9. The portion of the line $ax + by - 1 = 0$, intercepted between the lines $ax + y + 1 = 0$ and $x + by = 0$ subtends a right angle at the origin and the condition in a and b is $\lambda a + b + b^2 = 0$, then find value of λ .
10. If the straight lines joining the origin and the points of intersection of the curve $5x^2 - 24xy - 6y^2 + 4x - 2y + 3 = 0$ and $x + ky - 1 = 0$ are equally inclined to the x-axis, then find the value of $|k|$.
11. If the points of intersection of curves $C_1 = 4y^2 - 2\lambda x^2 - 2xy - 9x + 3$ and $C_2 = 2x^2 + 3y^2 - 4xy + 3x - 1$ subtends a right angle at origin, then find the value of λ .
12. Let the integral points inside or on the boundary of region bounded by straight lines as shown in figure is equal to k, then $\sqrt{k}$ is equal to





PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

- Point P(2, 3) lies on the line $4x + 3y = 17$. Then find the co-ordinates of points farthest from the line which are at 5 units distance from the P.
(A) (6, 6) (B) (6, -6) (C) (2, 0) (D) (-2, 0)
- Find the equation of the line passing through the point (2, 3) & making intercept of length 2 units between the lines $y + 2x = 3$ & $y + 2x = 5$.
(A) $3x - 4y = 18$ (B) $x = 2$ (C) $3x + 4y = 18$ (D) $x + 2 = 0$
- In a triangle ABC, co-ordinates of A are (1, 2) and the equations to the medians through B and C are $x + y = 5$ and $x = 4$ respectively. Then the co-ordinates of B and C will be
(A) (-2, 7), (4, 3) (B) (7, -2), (4, 3) (C) (2, 7), (-4, 3) (D) (2, -7), (3, -4)
- A is a point on either of two rays $y + \sqrt{3}|x| = 2$ at a distance of $\frac{4}{\sqrt{3}}$ units from their point of intersection. The co-ordinates of the foot of perpendicular from A on the bisector of the angle between them is/are
(A) $\left(-\frac{2}{\sqrt{3}}, 2\right)$ (B) (0, 0) (C) $\left(\frac{2}{\sqrt{3}}, 2\right)$ (D) (0, 4)
- If one side of a square is parallel to $3x - 4y = 0$ & its area being 16 while centre being (1, 1), then find equation of sides of square.
(A) $3x - 4y + 11 = 0$ (B) $3x - 4y - 9 = 0$ (C) $4x + 3y + 3 = 0$ (D) $4x + 3y - 17 = 0$
- Find the equations of the sides of a triangle having (4, -1) as a vertex, if the lines $x - 1 = 0$ and $x - y - 1 = 0$ are the equations of two internal bisectors of its angles.
(A) $2x - y + 3 = 0$ (B) $x + 2y - 6 = 0$ (C) $2x + y - 7 = 0$ (D) $x - 2y - 6 = 0$
- A straight line L with negative slope passes through the point (8, 2) and cuts the positive coordinate axes at points P and Q, then the correct statement(s) among the following is/are (O is origin)
(A) The absolute minimum value of $OP + OQ$, where O is origin is $18\sqrt{2}$
(B) Minimum area of $\triangle OPQ$ is 32
(C) The absolute minimum value of $OP + OQ$, where O is origin is 18
(D) Area of $\triangle OPQ$ is minimum for slope $\left(-\frac{1}{4}\right)$.
- The equation of the diagonals of a rectangle are $y + 8x - 17 = 0$ and $y - 8x + 7 = 0$. If the area of the rectangle is 8 sq. units, find the equation of the sides of the rectangle.
(A) $y = 1$ (B) $y = 9$ (C) $x = 1$ (D) $x = 2$.
- Two adjacent sides of a rhombus are $2x + 3y = a - 5$ and $3x + 2y = 4 - 2a$ and its diagonals intersect at the point (1, 2), then a can be -



(A) -16

(B) 16

(C) $-\frac{10}{3}$ (D) $\frac{10}{3}$

10. A line $L_1 \equiv 3y - 2x - 6 = 0$ is rotated about its point of intersection with y-axis in clockwise direction to make it L_2 such that the area formed by L_1 , L_2 , x-axis and line $x = 5$ is $\frac{49}{3}$ sq units if its point of intersection with $x = 5$ lies below x-axis then points lying on the equation of L_2 are
 (A) (3, -1) (B) (4,2) (C) (1,1) (D) (3,3)
11. Let $D(x_4, y_4)$ be a point such that ABCD is a square & M & P are the midpoints of the sides BC & CD respectively, then
 (A) Ratio of the areas of $\triangle AMP$ and the square is 3 : 8
 (B) Ratio of the areas of $\triangle MCP$ & $\triangle AMD$ is 1 : 1
 (C) Ratio of the areas of $\triangle ABM$ & $\triangle ADP$ is 1 : 1
 (D) Ratio of the areas of the quadrilateral AMCP and the square is 1 : 3
12. The equations of perpendicular of the sides AB & AC of $\triangle ABC$ are $x - y - 4 = 0$ and $2x - y - 5 = 0$ respectively. If the vertex A is $(-2, 3)$ and circumcenter is $\left(\frac{3}{2}, \frac{5}{2}\right)$, then which of the following is true.
 (A) equation of median of side AB is $x - y + 1 = 0$
 (B) centroid of triangle ABC is (3, 1)
 (C) vertex C is (4, 0)
 (D) Area of triangle ABC is 12.
13. Triangle ABC lies in the cartesian plane and has an area of 70 sq. units. The coordinates of B and C are (12, 19) and (23, 20) respectively and the coordinates of A are (p, q). The median to the side BC has slope -5, then which can be corrected.
 (A) $p + q = 47$ (B) $p + q = 27$ (C) $p - q = 17$ (D) $p - q = 13$
14. All the points lying on or inside the triangle formed by the points (1, 3), (5, 6) and (-1, 2) satisfy
 (A) $3x + 2y \geq 0$ (B) $2x + y + 1 \geq 0$ (C) $2x + 3y - 12 \geq 0$ (D) $2x + 11 \geq 0$
15. $A \equiv (4, 2)$ and $B \equiv (2, 4)$ are two given points and a point P on the line $3x + 2y + 10 = 0$ is given then which of the following is/are true.
 (A) (PA + PB) is minimum when $P\left(\frac{-14}{5}, \frac{-4}{5}\right)$ (B) (PA + PB) is maximum when $P\left(\frac{-14}{5}, \frac{-4}{5}\right)$
 (C) $|PA - PB|$ is maximum when $P(-22, 28)$ (D) (PA - PB) is minimum when $P(-22, 28)$.
16. A line passing through $P = (\sqrt{3}, 0)$ and making an angle of 60° with positive direction of x-axis cuts the parabola $y^2 = x + 2$ at A and B, then :
 (A) $PA + PB = \frac{2}{3}$ (B) $|PA - PB| = \frac{2}{3}$
 (C) $(PA)(PB) = \frac{4(2 + \sqrt{3})}{3}$ (D) $\frac{1}{PA} + \frac{1}{PB} = \frac{2 - \sqrt{3}}{2}$
17. Let $u \equiv ax + by + a\sqrt[3]{b} = 0$, $v \equiv bx - ay + b\sqrt[3]{b} = 0$, where $a, b \in \mathbb{R}$ be two straight lines, then find the equations of the bisectors of the angles formed by $k_1u - k_2v = 0$ & $k_1u + k_2v = 0$ for non zero real k_1 & k_2 are :
 (A) $u = 0$ (B) $k_2u + k_1v = 0$ (C) $k_2u - k_1v = 0$ (D) $v = 0$





18. The sides of a triangle are the straight line $x + y = 1$, $7y = x$ and $\sqrt{3}y + x = 0$. Then which of the following is an interior points of triangle ?
 (A) circumcentre (B) centroid (C) incentre (D) orthocentre
19. The line ' ℓ_1 ' passing through the point $(1, 1)$ and the ' ℓ_2 ' passes through the point $(-1, 1)$. If the difference of the slope of lines is 2. Find the locus of the point of intersection of the ℓ_1 and ℓ_2 .
 (A) $x^2 = y$ (B) $y = 2 - x^2$ (C) $y^2 = x$ (D) $x = 2 - y^2$
20. The two lines pairs $y^2 - 4y + 3 = 0$ and $x^2 + 4xy + 4y^2 - 5x - 10y + 4 = 0$ enclose a 4 sided convex polygon, then the correct statement among the following is/are
 (A) Area of polygon is 6 (B) Length of its diagonals are $\sqrt{5}$ & $\sqrt{53}$
 (C) Point of intersection of diagonals is $(-2, 2)$ (D) Polygon is parallelogram.
21. If the distance between the lines represented $9x^2 - 24xy + 16y^2 + k(6x - 8y) = 0$ is 4, then k may be
 (A) 3 (B) 10 (C) -10 (D) 7

PART - IV : COMPREHENSION

Comprehension # 1 (Q. NO. 1 to 3)

Let ABC be an acute angled triangle and AD, BE and CF are its medians, where E and F are the points $(3, 4)$ and $(1, 2)$ respectively and centroid of $\triangle ABC$ is $G(3, 2)$, then answer the following questions :

1. The equation of side AB is
 (A) $2x + y = 4$ (B) $x + y - 3 = 0$ (C) $4x - 2y = 0$ (D) none of these
2. Co-ordinates of D are
 (A) $(7, -4)$ (B) $(5, 0)$ (C) $(7, 4)$ (D) $(-3, 0)$
3. Height of altitude drawn from point A is (in units)
 (A) $4\sqrt{2}$ (B) $3\sqrt{2}$ (C) $6\sqrt{2}$ (D) $2\sqrt{3}$

Comprehension # 2 (Q.No. 4 to 6)

If vertices of triangle are $P(p_1, p_2)$, $Q(q_1, q_2)$, $R(r_1, r_2)$, then area of $\triangle PQR = \frac{1}{2} \begin{vmatrix} p_1 & p_2 & 1 \\ q_1 & q_2 & 1 \\ r_1 & r_1 & 1 \end{vmatrix}$ and if P, Q, R

are collinear, then $\begin{vmatrix} p_1 & p_2 & 1 \\ q_1 & q_2 & 1 \\ r_1 & r_1 & 1 \end{vmatrix} = 0$.

On the basis of above answer the following question.

4. If $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are the vertices of the triangle then find equation of median through A.

- (A) $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} - \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$ (B) $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$
- (C) $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + \begin{vmatrix} x & y & 1 \\ x_3 & y_3 & 1 \\ x_1 & y_1 & 1 \end{vmatrix} = 0$ (D) None of these





5. If $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are the vertices of the triangle then find equation of line through A and parallel to BC

(A) $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} - \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

(B) $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

(C) $\begin{vmatrix} x & y & 1 \\ x_2 & y_2 & 1 \\ x_1 & y_1 & 1 \end{vmatrix} + \begin{vmatrix} x & y & 1 \\ x_3 & y_3 & 1 \\ x_1 & y_1 & 1 \end{vmatrix} = 0$

(D) $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = \begin{vmatrix} x & y & 1 \\ x_3 & y_3 & 1 \\ x_1 & y_1 & 1 \end{vmatrix}$

6. If $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are the vertices of the triangle then find the equation of internal angle bisector through A

(A) $b \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} - c \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

(B) $c \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + b \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

(C) $b \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} + c \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

(D) $c \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} - b \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

COMPREHENSION # 3 (7-9)

The equation of pair of straight lines bisecting the angles between the straight lines

$ax^2 + 2hxy + by^2 = 0$ is $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$

The equation of pair of straight lines bisecting the angles between the straight lines

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is $\frac{(x - \alpha)^2 - (y - \beta)^2}{a - b} = \frac{(x - \alpha)(y - \beta)}{h}$ where (α, β) is point of intersecting of lines $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

7. Find the combined equation of the bisectors of the angle between the lines represented by $3x^2 - 5xy + 4y^2 = 0$
- (A) $5x^2 - 2xy - 5y^2 = 0$ (B) $5x^2 - 5y^2 + 4xy = 0$
 (C) $5x^2 - 5y^2 + 2xy = 0$ (D) $5x^2 - 5y^2 - xy = 0$
8. Find the combined equation of the bisectors of the angle between the lines represented by $2x^2 - 3y^2 + xy - x + 11y - 10 = 0$
- (A) $x^2 - y^2 + xy + 4x + 2 = 0$ (B) $x^2 - y^2 - 10xy + 8x + 12y - 10 = 0$
 (C) $x^2 - y^2 - 5xy + 3x + 7y - 5 = 0$ (D) None of these
9. Let combined equation of lines ℓ_1 and ℓ_2 is $x^2 - 2pxy - y^2 = 0$ combined equation of lines ℓ_3 and ℓ_4 is $x^2 - 2qxy - y^2 + 2gx + 2fy + c = 0$. If angle between lines (ℓ_1, ℓ_3) , (ℓ_1, ℓ_4) , (ℓ_2, ℓ_3) and (ℓ_2, ℓ_4) is equal to 45° , then find the value of pq .
- (A) -4 (B) 4 (C) -1 (D) 1



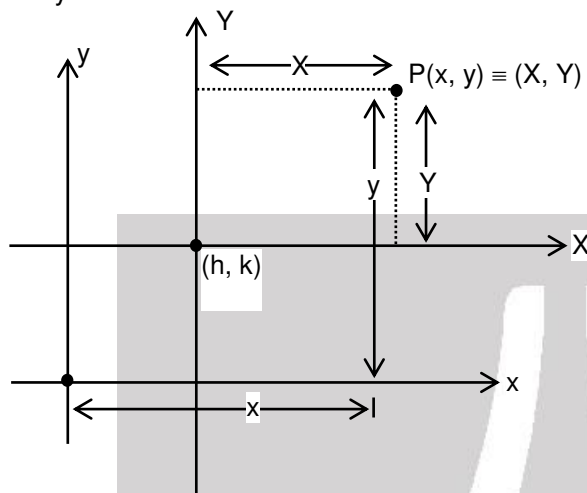


Comprehension (10 & 11)

Origin of coordinate system xy is shifted to (h, k) to make new coordinate system XY . X and Y are parallel to x and y . New co-ordinates of point $P(x, y)$ are $P(X, Y)$. x, y, X, Y are related as given below.

$$X = x - h$$

$$Y = y - k$$



10. Co-ordinates of $(-7, 9)$ if origin is shifted to $(2, 4)$ without changing direction of axes, are
 (A) $(-5, 13)$ (B) $(-7, 4)$ (C) $(-9, 5)$ (D) $(-9, 13)$
11. If co-ordinate axes are so translated such that ordinate of $(4, 12)$ becomes zero while abscissa remains same. Then new coordinates of point $(-8, -2)$ are
 (A) $(-8, 14)$ (B) $(-8, 10)$ (C) $(-8, -14)$ (D) $(-8, -10)$

Exercise-3

PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

* Marked Questions may have more than one correct option.

1. Let $O(0, 0)$, $P(3, 4)$, $Q(6, 0)$ be the vertices of the triangle OPQ . The point R inside the triangle OPQ is such that the triangles OPR , PQR , OQR are of equal area. The co-ordinates of R are

[IIT-JEE - 2007, P-II, (3, -1), 81]

- (A) $\left(\frac{4}{3}, 3\right)$ (B) $\left(3, \frac{2}{3}\right)$ (C) $\left(3, \frac{4}{3}\right)$ (D) $\left(\frac{4}{3}, \frac{2}{3}\right)$

2. Lines $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q , respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R .

[IIT-JEE - 2007, P-II, (3, -1), 81]

STATEMENT - 1 : The ratio $PR : RQ$ equals $2\sqrt{2} : \sqrt{5}$.

because

STATEMENT - 2 : In any triangle, bisector of an angle divides the triangle into two similar triangles.

- (A) Statement - 1 is True, Statement - 2 is True; Statement-2 **is** a correct explanation for Statement - 1
 (B) Statement - 1 is True, Statement - 2 is True; Statement-2 is **NOT** a correct explanation for Statement - 1
 (C) Statement - 1 is True, Statement - 2 is False
 (D) Statement - 1 is False, Statement - 2 is True



3. Consider three points [IIT-JEE - 2008, P-II, (3, -1), 81]
 $P = (-\sin(\beta - \alpha), -\cos \beta)$, $Q = (\cos(\beta - \alpha), \sin \beta)$ and $R = (\cos(\beta - \alpha + \theta), \sin(\beta - \theta))$, where
 $0 < \alpha, \beta, \theta < \frac{\pi}{4}$. Then,
 (A) P lies on the line segment RQ (B) Q lies on the line segment PR
 (C) R lies on the line segment QP (D) P, Q, R are non-collinear
4. The locus of the orthocentre of the triangle formed by the lines [IIT-JEE - 2009, Paper-2, (3, -1), 80]
 $(1 + p)x - py + p(1 + p) = 0$, $(1 + q)x - qy + q(1 + q) = 0$ and $y = 0$, where $p \neq q$, is
 (A) a hyperbola (B) a parabola (C) an ellipse (D) a straight line
5. A straight line L through the point (3, -2) is inclined at an angle 60° to the line $\sqrt{3}x + y = 1$. If L also intersects the x-axis, then the equation of L is [IIT-JEE 2011, Paper-1, (3, -1), 80]
 (A) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$ (B) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$
 (C) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$ (D) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$
6. For $a > b > c > 0$, the distance between (1, 1) and the point of intersection of the lines $ax + by + c = 0$ and $bx + ay + c = 0$ is less than $2\sqrt{2}$. Then [JEE (Advanced) 2013, Paper-1, (2, 0)/60]
 (A) $a + b - c > 0$ (B) $a - b + c < 0$ (C) $a - b + c > 0$ (D) $a + b - c < 0$
7. For a point P in the plane, let $d_1(P)$ and $d_2(P)$ be the distance of the point P from the lines $x - y = 0$ and $x + y = 0$ respectively. The area of the region R consisting of all points P lying in the first quadrant of the plane and satisfying $2 \leq d_1(P) + d_2(P) \leq 4$, is [JEE (Advanced) 2014, Paper-1, (3, 0)/60]

PART - II : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

1. The line L given by $\frac{x}{5} + \frac{y}{b} = 1$ passes through the point (13, 32). The line K is parallel to L and has the equation $\frac{x}{c} + \frac{y}{3} = 1$. Then the distance between L and K is [AIEEE - 2010 (8, -2), 144]
 (1) $\sqrt{17}$ (2) $\frac{17}{\sqrt{15}}$ (3) $\frac{23}{\sqrt{17}}$ (4) $\frac{23}{\sqrt{15}}$
2. The line $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R. [AIEEE - 2011, I(4, -1), 120]
Statement-1 : The ratio PR : RQ equals $2\sqrt{2} : \sqrt{5}$
Statement-2 : In any triangle, bisector of an angle divides the triangle into two similar triangles.
 (1) Statement-1 is true, Statement-2 is true ; Statement-2 is correct explanation for Statement-1
 (2) Statement-1 is true, Statement-2 is true ; Statement-2 is **not** a correct explanation for Statement-1
 (3) Statement-1 is true, Statement-2 is false
 (4) Statement-1 is false, Statement-2 is true
3. The lines $x + y = |a|$ and $ax - y = 1$ intersect each other in the first quadrant. Then the set of all possible values of a is the interval : [AIEEE - 2011, II(4, -1), 120]
 (1) $(0, \infty)$ (2) $[1, \infty)$ (3) $(-1, \infty)$ (4) $(-1, 1]$
4. If A(2, -3) and B(-2, 1) are two vertices of a triangle and third vertex moves on the line $2x + 3y = 9$, then the locus of the centroid of the triangle is : [AIEEE - 2011, II(4, -1), 120]
 (1) $x - y = 1$ (2) $2x + 3y = 1$ (3) $2x + 3y = 3$ (4) $2x - 3y = 1$
5. If the line $2x + y = k$ passes through the point which divides the line segment joining the points (1, 1) and (2, 4) in the ratio 3 : 2, then k equals : [AIEEE-2012, (4, -1)/120]
 (1) $\frac{29}{5}$ (2) 5 (3) 6 (4) $\frac{11}{5}$





6. A line is drawn through the point (1, 2) to meet the coordinate axes at P and Q such that it forms a triangle OPQ, where O is the origin. If the area of the triangle OPQ is least, then the slope of the line PQ is : **[AIEEE-2012, (4, -1)/120]**
 (1) $-\frac{1}{4}$ (2) -4 (3) -2 (4) $-\frac{1}{2}$
7. A ray of light along $x + \sqrt{3}y = \sqrt{3}$ gets reflected upon reaching x-axis, the equation of the reflected ray is **[AIEEE - 2013, (4, -1), 360]**
 (1) $y = x + \sqrt{3}$ (2) $\sqrt{3}y = x - \sqrt{3}$ (3) $y = \sqrt{3}x - \sqrt{3}$ (4) $\sqrt{3}y = x - 1$
8. The x-coordinate of the incentre of the triangle that has the coordinates of mid points of its sides as (0, 1), (1, 1) and (1, 0) is : **[AIEEE - 2013, (4, -1), 360]**
 (1) $2 + \sqrt{2}$ (2) $2 - \sqrt{2}$ (3) $1 + \sqrt{2}$ (4) $1 - \sqrt{2}$ **[AIEEE - 2013, (4, -1), 360]**
9. Let PS be the median of the triangle with vertices P(2, 2), Q (6, -1), and R (7, 3). The equation of the line passing through (1, -1) and parallel to PS is : **[JEE(Main) 2014, (4, -1), 120]**
 (1) $4x + 7y + 3 = 0$ (2) $2x - 9y - 11 = 0$ (3) $4x - 7y - 11 = 0$ (4) $2x + 9y + 7 = 0$
10. Let a, b, c and d be non-zero numbers. If the point of intersection of the lines $4ax + 2ay + c = 0$ and $5bx + 2by + d = 0$ lies in the fourth quadrant and is equidistant from the two axes then : **[JEE(Main) 2014, (4, -1), 120]**
 (1) $3bc - 2ad = 0$ (2) $3bc + 2ad = 0$ (3) $2bc - 3ad = 0$ (4) $2bc + 3ad = 0$
11. The number of points, having both co-ordinates as integers, that lie in the interior of the triangle with vertices (0, 0), (0, 41) and (41, 0) is **[JEE(Main) 2015, (4, -1), 120]**
 (1) 901 (2) 861 (3) 820 (4) 780
12. Two sides of a rhombus are along the lines, $x - y + 1 = 0$ and $7x - y - 5 = 0$. If its diagonals intersect at (-1, -2), then which one of the following is a vertex of this rhombus ? **[JEE(Main) 2016, (4, -1), 120]**
 (1) (-3, -8) (2) $\left(\frac{1}{3}, -\frac{8}{3}\right)$ (3) $\left(-\frac{10}{3}, -\frac{7}{3}\right)$ (4) (-3, -9)
13. Let k be an integer such that the triangle with vertices (k, -3k), (5, k) and (-k, 2) has area 28 sq. units. Then the orthocentre of this triangle is at the point : **[JEE(Main) 2017, (4, -1), 120]**
 (1) $\left(2, -\frac{1}{2}\right)$ (2) $\left(1, \frac{3}{4}\right)$ (3) $\left(1, -\frac{3}{4}\right)$ (4) $\left(2, \frac{1}{2}\right)$
14. A straight line through a fixed point (2,3) intersects the coordinate axes at distinct points P and Q. If O is the origin and the rectangle OPRQ is completed, then the locus of R is **[JEE(Main) 2018, (4, -1), 120]**
 (1) $3x + 2y = xy$ (2) $3x + 2y = 6xy$ (3) $3x + 2y = 6$ (4) $2x + 3y = xy$
15. Consider the set all lines $px + qy + r = 0$ such that $3p + 2q + 4r = 0$. Which one of the following statements is true ? **[JEE(Main) 2019, Online (09-01-19), P-1 (4, -1), 120]**
 (1) The lines are not concurrent (2) The lines are all parallel
 (3) The lines are concurrent at the point $\left(\frac{3}{4}, \frac{1}{2}\right)$ (4) Each line passes through the origin.





16. Let the equations of two sides of a triangle be $3x - 2y + 6 = 0$ and $4x + 5y - 20 = 0$. If the orthocentre of this triangle is at $(1, 1)$, then the equation of its third side is :
[JEE(Main) 2019, Online (09-01-19), P-2 (4, - 1), 120]
 (1) $26x - 122y - 1675 = 0$ (2) $26x + 61y + 1675 = 0$
 (3) $122y - 26x - 1675 = 0$ (4) $122y + 26x + 1675 = 0$
17. Two vertices of a triangle are $(0, 2)$ and $(4, 3)$. If its orthocentre is at the origin, then its third vertex lies in which quadrant ?
[JEE(Main) 2019, Online (10-01-19), P-2 (4, - 1), 120]
 (1) third (2) second (3) first (4) fourth
18. Let $O(0,0)$ and $A(0,1)$ be two fixed points. Then the locus of a point P such that the perimeter of $\triangle AOP$ is 4, is :
 (1) $9x^2 - 8y^2 + 8y = 16$ (2) $8x^2 - 9y^2 + 9y = 18$
 (3) $9x^2 + 8y^2 - 8y = 16$ (4) $8x^2 + 9y^2 - 9y = 18$
[JEE(Main) 2019, Online (08-04-19), P-1 (4, - 1), 120]
19. A straight line L at a distance of 4 units from the origin makes positive intercepts on the coordinate axes and the perpendicular from the origin to this line makes an angle of 60° with the line $x + y = 0$. Then an equation of the line L is -
 (1) $(\sqrt{3} + 1)x + (\sqrt{3} - 1)y = 8\sqrt{2}$ (2) $(\sqrt{3} - 1)x + (\sqrt{3} + 1)y = 8\sqrt{2}$
 (3) $\sqrt{3}x + y = 8$ (4) $x + \sqrt{3}y = 8$
[JEE(Main) 2019, Online (12-04-19), P-2 (4, - 1), 120]
20. The locus of the mid-point of the perpendiculars drawn from points on the line, $x = 2y$ to the line $x = y$ is:
 (1) $3x - 2y = 0$ (2) $3x - 3y = 0$ (3) $5x - 7y = 0$ (4) $7x - 5y = 0$
[JEE(Main) 2020, Online (07-01-20), P-2 (4, - 1), 120]





Answers

EXERCISE - 1

PART - I

Section (A):

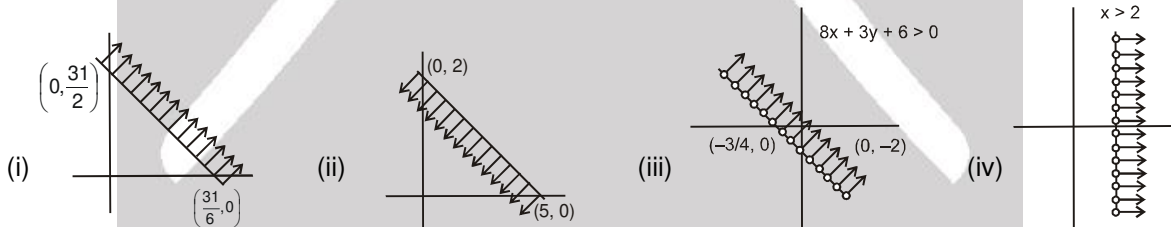
- A-1.** (ii) (3, 4) (6, 8) **A-2.** (i) 1:3 internally (ii) 1:2
A-3. (i) $\frac{3}{2}$ (ii) 4 : 1 internally (iii) (8, -8) **A-4.** (7, 2) or (1, 0) **A-5.** (i) 15 sq. units (ii) $|ab|$ sq. units

Section (B) :

- B-1.** (i) $y = -\frac{1}{\sqrt{3}}x - \frac{4}{\sqrt{3}}$, slope = $-\frac{1}{\sqrt{3}}$, y-intercept = $-\frac{4}{\sqrt{3}}$
 (ii) $\frac{x}{-4} + \frac{y}{-4/\sqrt{3}} = 1$, x-intercept = -4, y-intercept = $-\frac{4}{\sqrt{3}}$
 (iii) $x \cos 240^\circ + y \sin 240^\circ = 2$, $P = 2$, $\alpha = 240^\circ$ **B-2.** 3 **B-3.** $8x + 5y + 60 = 0, 2x + 5y + 30 = 0$
B-4. Equations are $2x - 3y + 15 = 0, 2x + y - 5 = 0, 6x - y - 27 = 0$
 Length of sides are $2\sqrt{13}, 4\sqrt{5}, 2\sqrt{37}$
B-5. $x - y + 1 = 0$ **B-6.** $x + y = 5, x - 7y + 3 = 0$ **B-7.** 14 **B-8.** $\pi/12, 5\pi/12$
B-9. $x = 3, y = 4, 9/2$ sq. units **B-10.** $x - 5y + 3 = 0, 5x + y - 11 = 0$ **B-11.** $(-6, 9 - 2\sqrt{3})$
B-12. $x^2 + y^2 - 6x - 6y = 0$. **B-13.** 18 **B-14.** $7x - 4y + 25 = 0, 7x - 4y - 3 = 0, 4x + 7y = 11$

Section (C) :

C-1.



- C-2.** Foot $\left(\frac{23}{10}, \frac{29}{10}\right)$, Image $\left(\frac{13}{5}, \frac{14}{5}\right)$, $x + 3y - 11 = 0$ **C-3.** $\frac{1+\sqrt{10}}{3}$ **C-4.** $\left(\frac{3}{25}, \frac{4}{25}\right)$
C-5. Inside **C-6.** 3 sq. unit **C-7.** 2 **C-8.** $x + 7y + 12 = 0, x + 7y - 8 = 0$

Section (D) :

- D-1.** (i) (7, 8) (ii) $\left(\frac{21}{2}, \frac{8}{3}\right)$ (iii) (7, 9) (iv) (27, -21) **D-2.** 7
D-3. $2x + 3y + 8 = 0, 4x - 3y - 7 = 0, 8x + 3y + 9 = 0$, orthocentre $\left(-\frac{1}{6}, -\frac{23}{9}\right)$ **D-4.** (0, 0)
D-6. $\frac{3}{x} + \frac{2}{y} = 3$ **D-7.** $(3x - 1)^2 + 9y^2 = a^2 + b^2$ **D-9.** $x^2 + y^2 = a^2 + b^2$ **D-10.** $y = 2x + 1$ or $y = -2x + 1$

Section (E) :

- E-1.** acute $2x + y - 3 = 0$, obtuse $x - 2y + 1 = 0$, origin lies in obtuse angle bisector.
E-2. $9x - 7y = 1, 7x + 9y = 73$ **E-3.** $5x + 5y - 3 = 0$ **E-4.** $\frac{6}{5}, \frac{1}{2}, -3$





E-5. 0 **E-6.** (i) $43x - 29y = 71$ (ii) $x + y + 2 = 0$

E-7. $2xy(ma-b\ell) + x(an-\ell c) + y(mc-bn) = 0$.

Section (F) :

F-2. $\lambda = 2$, (i) $3x - y + 2 = 0$, $4x - 2y + 1 = 0$, (ii) point of intersection $\left(-\frac{3}{2}, -\frac{5}{2}\right)$, (iii) $\tan^{-1}\left(\frac{1}{7}\right)$,

F-3. (i) $h = 1$ (ii) $k = 1$ **F-4.** $4ax^2 + 4amxy + ny^2 = 0$; $4a\ell + n = 0$

F-5. $a^2 + b^2 = (a')^2 + (b')^2 \Rightarrow (a + a')x + (b + b')y + c + c' = 0 \Rightarrow (a - a')x + (b - b')y = 0$

PART - II

Section (A):

A-1. (C) **A-2.** (B) **A-3.** (B) **A-4.** (A) **A-5.** (A)

Section (B) :

B-1. (D) **B-2.** (B) **B-3.** (C) **B-4.** (B) **B-5.** (D) **B-6.** (B) **B-7.** (B)

B-8. (B) **B-9.** (A) **B-10.** (B) **B-11.** (B) **B-12.** (C) **B-13.** (B)

Section (C) :

C-1. (B) **C-2.** (B) **C-3.** (A) **C-4.** (C) **C-5.** (A) **C-6.** (C)

Section (D) :

D-1. (A) **D-2.** (D) **D-3.** (C) **D-4.** (B) **D-5.** (B) **D-6.** (A) **D-7.** (A)

D-8. (A)

Section (E) :

E-1. (A) **E-2.** (C) **E-3.** (A) **E-4.** (D) **E-5.** (D) **E-6.** (B) **E-7.** (D)

E-8. (B) **E-9.** (C)

Section (F) :

F-1. (D) **F-2.** (C) **F-3.** (A)

PART - III

1. (A) $\rightarrow$ (q, s) (B) $\rightarrow$ (r, s) (C) $\rightarrow$ (r, s) (D) $\rightarrow$ (r, s) 2. (A) $\rightarrow$ (p), (B) $\rightarrow$ (q), (C) $\rightarrow$ (r), (D) $\rightarrow$ (s)

EXERCISE - 2

PART - I

1. (C) 2. (C) 3. (D) 4. (A) 5. (D) 6. (C) 7. (B)
8. (A) 9. (C) 10. (A) 11. (A) 12. (C) 13. (C) 14. (A)
15. (B) 16. (A) 17. (B) 18. (A) 19. (B) 20. (A) 21. (A)

PART - II

1. 00.00 2. 02.98 3. 18.00 4. 01.00 5. 33.80 6. 04.72 or 04.73
7. 04.00 8. 04.19 or 04.20 9. 02.00 10. 02.60 11. 09.50 12. 18.19

PART - III

1. (AD) 2. (BC) 3. (B) 4. (B) 5. (ABCD) 6. (ACD) 7. (BCD)
8. (ABCD) 9. (AD) 10. (AC) 11. (AC) 12. (C) 13. (ABD) 14. (ABD)
15. (AC) 16. (BC) 17. (AD) 18. (BC) 19. (AB) 20. (ABD) 21. (BC)

PART - IV

1. (A) 2. (B) 3. (C) 4. (B) 5. (A) 6. (C) 7. (B)
5. (A) 6. (C) 7. (A) 8. (D) 9. (C) 10. (C) 11. (C)

EXERCISE - 3

PART - I

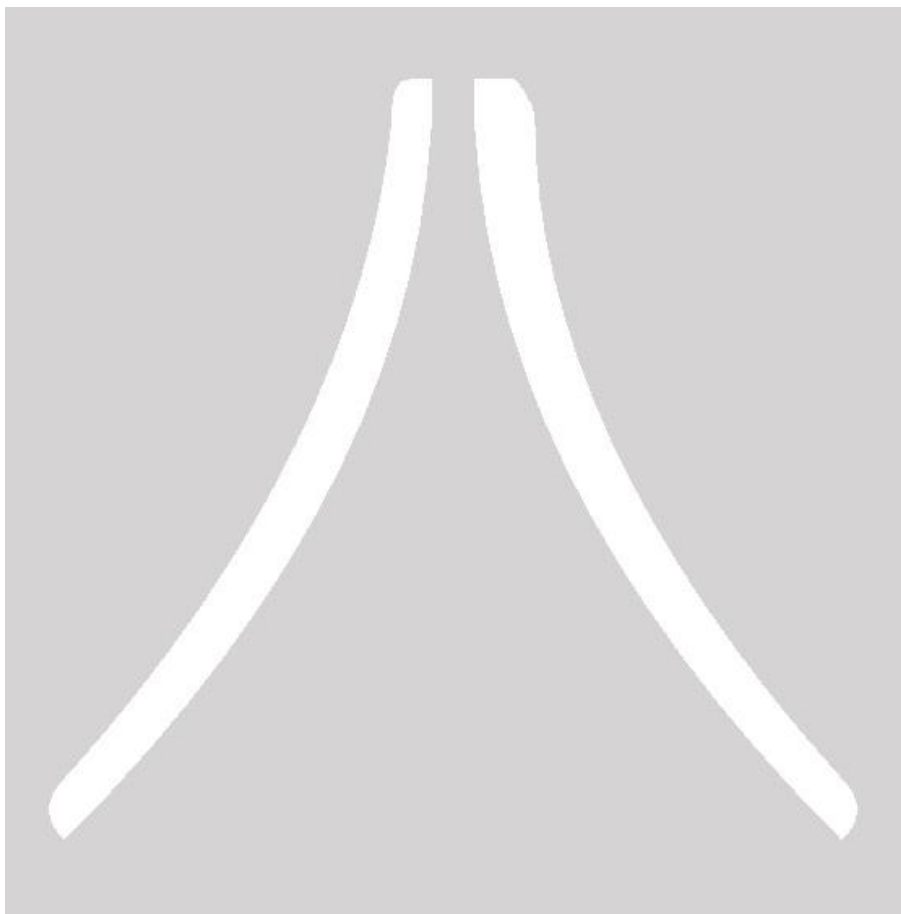
1. (C) 2. (C) 3. (D) 4. (D) 5. (B) 6. (A) or (C) or Bonus
7. (6)

PART - II

1. (3) 2. (3) 3. (2) 4. (2) 5. (3) 6. (3) 7. (2)
8. (2) 9. (4) 10. (1) 11. (4) 12. (2) 13. (4) 14. (1)



15. (3) 16. (1) 17. (2) 18. (3) 19. (2) 20. (3)





High Level Problems (HLP)

Marked Questions may have for Revision Questions.

- The vertices of a triangle OBC are O(0,0) B(-3,-1) and C(-1,-3). Find the equation of line parallel to BC and intersecting the sides OB and OC, whose perpendicular distance from the point (0,0) is $\frac{1}{2}$.
- A variable line, drawn through the point of intersection of the straight lines $\frac{x}{a} + \frac{y}{b} = 1$ and $\frac{x}{b} + \frac{y}{a} = 1$, meets the coordinate axes in A & B. Show that the locus of the mid point of AB is the curve $2xy(a+b) = ab(x+y)$.
- From the vertices A, B, C of a triangle ABC, perpendiculars AD, BE, CF are drawn to any straight line. Show that the perpendiculars from D, E, F to BC, CA, AB respectively are concurrent.
- A triangle is formed by the lines whose equations are AB : $x + y - 5 = 0$, BC : $x + 7y - 7 = 0$ and CA : $7x + y + 14 = 0$. Find the bisector of the interior angle at B and the exterior angle at C. Determine the nature of the interior angle at A and find the equation of the bisector.
- The coordinates of the feet of $\perp$ from the vertices of a Δ on the opposite sides are (20, 25), (8, 16) and (8, 9). Then find the coordinates of a vertices of the Δ .
- Let P is any point inside the triangle ABC of side lengths 6, 5, 5 units and p_1, p_2, p_3 be the lengths of perpendiculars drawn from P to the sides of triangle. Find the maximum value of $p_1.p_2.p_3$.
- Let in ΔPAB , A is (0, 0), B is (a, 0) and P is variable such that $\angle PBA$ is equal to three times $\angle PAB$ the, find the locus of P.
- Through a fixed point any straight line is drawn meeting two given parallel straight lines in P and Q, through P and Q straight lines are drawn in fixed directions, meeting in R. Prove that the locus of R is straight line.
- Through the origin O a straight line is drawn to cut the lines $y = m_1x + C_1$ and $y = m_2x + C_2$ at Q and R respectively. Find the locus of the point P on this variable line, such that OP is the geometric mean of OQ and OR.
- The sides of a triangle are $L_r \equiv x \cos \alpha_r + y \sin \alpha_r - p_r = 0$ for $r = 1, 2, 3$. Show that its orthocentre is given by $L_1 \cos (\alpha_2 - \alpha_3) = L_2 \cos (\alpha_3 - \alpha_1) = L_3 \cos (\alpha_1 - \alpha_2)$.
- A line passes through a fixed point R intersecting a fixed line at P. A point Q on RP such that $\frac{RP}{RQ}$ is constant. Then show that locus of Q is a straight line.
- A triangle ABC with $a = 8$, $b = 6$ and $c = 10$ slides on the coordinate axes with vertices A and B on the x-axis and the y-axis respectively. Find the locus of the vertex C.
- The line $L_1 \equiv 4x + 3y - 12 = 0$ intersects the x and the y-axis at A and B respectively. A variable line perpendicular to L_1 intersects the x and the y-axis at P and Q respectively. Find the locus of the circumcentre of triangle ABQ.
- Show that the orthocentre of Δ formed by the straight lines, $ax^2 + 2hxy + by^2 = 0$ and $\ell x + my = 1$ is a point (x', y') such that $\frac{x'}{\ell} = \frac{y'}{m} = \frac{a+b}{am^2 - 2h\ell m + b\ell^2}$.

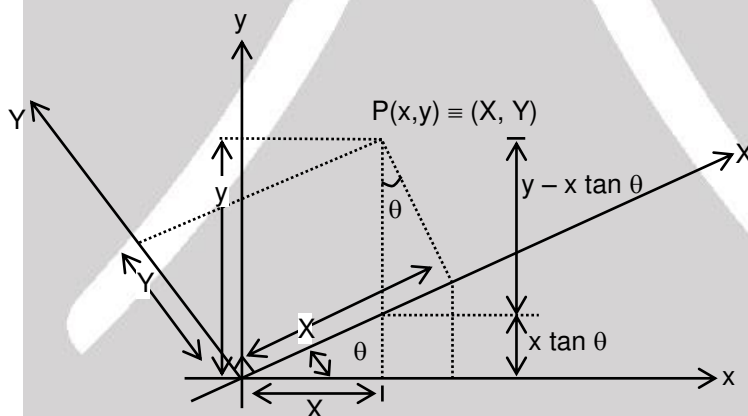




15. The distance of a point (x_1, y_1) from each of two straight lines which passes through the origin of co-ordinates is δ , find the combined equation of these straight lines.
16. A point moves so that the distance between the feet of the perpendiculars from it on the lines $ax^2 + 2hxy + by^2 = 0$ is a constant $2d$. Show that the equation to its locus is, $(x^2 + y^2)(h^2 - ab) = d^2 \{(a - b)^2 + 4h^2\}$.
17. Show that the pair of lines given by $a^2 x^2 + 2h(a + b)xy + b^2 y^2 = 0$ is equally inclined to the pair given by $ax^2 + 2hxy + by^2 = 0$.
18. All the chords of the curve $3x^2 - y^2 - 2x + 4y = 0$ which subtend a right angle at the origin are concurrent. Does this result also hold for the curve, $3x^2 + 3y^2 - 2x + 4y = 0$? If yes, what is the point of concurrence.
19. The straight lines $(A^2 - 3B^2)x^2 + 8ABxy + (B^2 - 3A^2)y^2 = 0$ form a Δ with the line $Ax + By + C = 0$, then prove that
 - (i) Area of Δ is $\frac{C^2}{\sqrt{3}(A^2 + B^2)}$
 - (ii) Δ is equilateral
 - (iii) The orthocentre of Δ does not lie on one of its vertices

Comprehension (Q. 20 & 21)

If coordinate system xy is being rotated through an angle θ in anti clock wise direction about the origin as shown in the diagram, Coordinates of $P(x, y)$ has been change to $P(X, Y)$ in new coordinate system XY , then x, y, X, Y are related as given below.

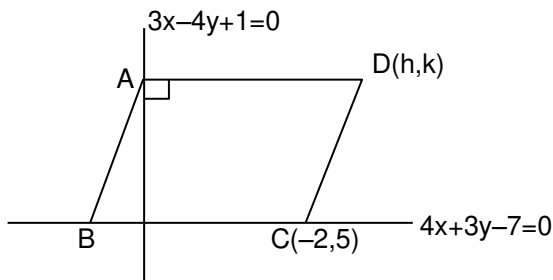


$$\begin{aligned}
 X &= x \sec \theta + (y - x \tan \theta) \sin \theta & \text{and} & & Y &= (y - x \tan \theta) \cos \theta \\
 &= x \sec \theta + y \sin \theta - \frac{x \sin^2 \theta}{\cos \theta} & & & &= y \cos \theta - x \sin \theta \\
 \boxed{X} &= \boxed{x \cos \theta + y \sin \theta} & & & \boxed{Y} &= \boxed{-x \sin \theta + y \cos \theta}
 \end{aligned}$$

20. If the axes are rotated through 60° in anticlockwise direction about origin. Find co-ordinates of point $(2, 6)$ in new co-ordinate axes.
21. If axes are rotated through an acute angle in clockwise direction about origin so that equation $x^2 + 2xy + y^2 - 2x + 2y = 0$ becomes free from xy in its new position, then find equation in new position
22. Find the acute angle between two straight lines passing through the point $M(-6, -8)$ and the points in which the line segment $2x + y + 10 = 0$ enclosed between the co-ordinate axes is divided in the ratio $1 : 2 : 2$ in the direction from the point of its intersection with the x -axis to the point of intersection with the y -axis.



- 23_. Let A lies on $3x - 4y + 1 = 0$. B lies on $4x + 3y - 7 = 0$ and C is $(-2, 5)$. If ABCD is rhombus, then find locus of D.



$$DA = DC$$

$$(h + 2)^2 + (A - 5)^2 = \left(\frac{3h - 4k + 1}{5} \right)^2$$

$$25((x + 2)^2 + (y - 5)^2) = (3x - 4y + 1)^2$$

- 24_. Let D is point on line $\ell_1 : x + y - 2 = 0$ and $S(3, 3)$ is fixed point. ℓ_2 is the line perpendicular to DS and passing through S. If M is another point on line ℓ_1 (other than D), then find locus of point of intersection of ℓ_2 and angle bisector of $\angle MDS$.
- 25_. A variable line cuts the line $2y = x - 2$ and $2y = -x + 2$ in points A and B respectively. If A lies in first quadrant, B lies in 4th quadrant and area of $\triangle AOB$ is 4, then find locus of
(i) mid point of AB (ii) centroid of $\triangle OAB$
26. An equilateral triangle PQR is formed where P $(1, 3)$ is fixed point and Q is moving point on line $x = 3$. Find the locus of R.

Answers

1. $x + y + \frac{1}{\sqrt{2}} = 0$. 4. $3x + 6y - 16 = 0; 8x + 8y + 7 = 0; 12x + 6y - 11 = 0$
5. $(5, 10), (50, -5), (15, 30)$ 6. $p_1 p_2 p_3 \leq \frac{256}{75}$ 7. $4x^3 - 4xy^2 - 3ax^2 + ay^2 = 0$.
9. $(y - m_1 x)(y - m_2 x) = c_1 c_2$ 12. $\frac{y}{x} = \frac{3}{4}$ or $3x - 4y = 0$ 13. $6x - 8y + 7 = 0$
15. $(y_1^2 - \delta^2)x^2 - 2x_1 y_1 xy + (x_1^2 - \delta^2)y^2 = 0$ 18. $(1, -2), \text{ yes } (1/3, -2/3)$
20. $(1 + 3\sqrt{3}, -\sqrt{3} + 3)$ 21. $x^2 + \sqrt{2}y = 0$ 22. $\pi/4$
23. $25((x + 2)^2 + (y - 5)^2) = (3x - 4y + 1)^2$ 24. $(x - 3)^2 + (y - 3)^2 = \left(\frac{x + y - 2}{\sqrt{2}} \right)^2$
25. (i) $(x - 1)^2 - 4y^2 = 9$ (ii) $(x - 2/3)^2 + 4y^2 = 4$ 26. $(x - 2) = \pm \sqrt{3}(y - 3 \pm \sqrt{3})$

